

Each of the two input files is provided as a parameter after the `-i` option.

- `-lavfi`: We need a background video stream so that we can overlay `video1` and `video2` on it. For this, we are using the libavfilter input virtual device (<https://www.ffmpeg.org/ffmpeg-all.html#lavfi>).
- `-filter_complex`: A filter complex is used to represent a complex filter graphs in FFmpeg. The parameter after the `-filter_complex` option (in double quotes) consists of a series of complex filter graphs separated by semicolons. Complex filter graphs are those which cannot be described simply as a linear processing chain applied to one stream. This is the case, for example, when the graph has more than one input and/or output, or when the output stream type is different from the input (see <https://www.ffmpeg.org/ffmpeg-all.html#Complex-filtergraphs>).

A complex filter graph is represented by a string of the form: `[in_link_1]...[in_link_N]filter_name=arguments[out_link_1]...[out_link_M]`. `in_link_x` represents the inputs that will be filtered using the `filter_name` and `out_link_x` represents the outputs. Multiple filters are separated by commas. Now, let us look at the filter graphs used in the command

```
[0:v]fade=t=out:st=5:d=1:alpha=1,setpts=PTS-STARTPTS[v0]
```

`[0:v]` represents the input video stream. The number to the left of `:` represents the input number (starting with 0) based on the order in which they are provided with the `-i` parameter. (So, 0 represents `video1.avi`, 1 represents `video2.avi`, etc). `'v'` to the right of `:` represents the video stream from `video1.avi`. We apply two filters to this input — `fade` and `setpts`. Since we want the first video to fade out and the second video to fade in at the point where they are stitched, we use the fade filter present in FFmpeg (<https://www.ffmpeg.org/ffmpeg-all.html#fade>). We set the fade type to `'out'` and the duration of fading to 1s, starting at 4s. (Remember that the `video1.avi` was of duration 5s. So, we want to fade out the video in the last 1s.) `setpts` is used to set the presentation time stamp in the output stream. `PTS` and `STARTPTS` are constants defined in FFmpeg (https://www.ffmpeg.org/ffmpeg-all.html#setpts_002c-asetpts). This filter will offset the presentation time stamp of all the frames in the output stream by the `PTS` of the first frame in the input stream. The output video stream is represented by `v0`.

```
[1:v]fade=t=in:st=0:d=1:alpha=1,setpts=PTS-STARTPTS+5/TB[v1]
```

In the above command, the video stream in the second input video (`video2.avi`) is the input video stream. The video will fade in for a duration of 1s, starting from the beginning. Now, this video has to be shown after the `video1.avi` has faded out (i.e., 5s). So, we use the `setpts` filter to offset the presentation time stamp by 5s. `TB` is a constant representing the time-base of input time stamps.

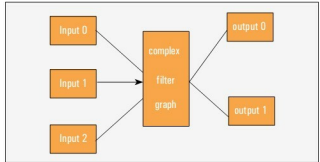


Figure 1: Representation of a complex filter graph

The output video stream is represented by `val1`.

```
[2:v]scale=1920x1080,trim=duration=15[base]
```

Here, the third video input (from the virtual device) acts as the input video stream. This stream is scaled to the resolution matching that of `video1.avi` and `video2.avi`. The stream is generated for a total duration of 15s and the output stream is represented by `base`.

```
[base][v0]over lay[base1];[base1][v1]over lay[outv]
```

The command given above will simply overlay the streams `v0` and `v1` on `base`. The resulting stream is represented as `outv`.

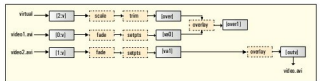


Figure 2: Various streams and filters used in the example command

- `-map`: The final video stream `outv` is designated as the source for an output file using this option (see <https://www.ffmpeg.org/ffmpeg-all.html#Advanced-options>)

Look at the command again. Hopefully it is not as scary now.

Adding an image

Often, there is a need to show a picture for a couple of seconds in a video (for example, when showing a logo). The command below will create a video (`img.avi`) that will show the image (`img.jpg`). The option `'-t'` sets the duration of the video to be 5s and the option `'-s'` sets the resolution.

```
ffmpeg -loop 1 -i img.jpg -t 5 -s 1920x1080 img.avi
```

Multiple videos

When there are more than two videos to be stitched, the procedure can be extended in a trivial manner. Though it